$$g(x) = \begin{cases} \frac{x^2-25}{x+5} & \text{if } x \neq -5 \\ \lambda & \text{if } x = -5 \end{cases}$$ Find λ such that $f(x) = g(x)$ for all $x \in R$

Solution

Given that $f(x) = g(x)$ for all $x \in R$

$\therefore \quad f(-5) = g(-5)$

$-5 - 5 = \lambda$

$\therefore \quad \lambda = -10$

Example 5.5

If $f(x) = 2^x$, then show that $f(x) \cdot f(y) = f(x+y)$

Solution

$f(x) = 2^x$ (Given)

$\therefore \quad f(x+y) = 2^{x+y}$

$= 2^x \cdot 2^y$

$= f(x) \cdot f(y)$

Example 5.6

If $f(x) = \frac{x-1}{x+1}$, $x > 0$, then show that $f[f(x)] = -\frac{1}{x}$

Solution

$$f(x) = \frac{x-1}{x+1} \text{ (Given)}$$

$$\therefore f[f(x)] = \frac{f(x)-1}{f(x)+1} = \frac{\frac{x-1}{x+1}-1}{\frac{x-1}{x+1}+1}$$

$$= \frac{x-1-x-1}{x-1+x+1} = -\frac{2}{2x} = -\frac{1}{x}$$

Example 5.7

If $f(x) = \log\frac{1+x}{1-x}$, $0 < x < 1$, then show that $f\left(\frac{2x}{1+x^2}\right) = 2f(x)$.

Solution

$$f(x) = \log\frac{1+x}{1-x} \text{ (Given)}$$

$$\therefore f\left(\frac{2x}{1+x^2}\right) = \log\frac{1+\frac{2x}{1+x^2}}{1-\frac{2x}{1+x^2}}$$

$$= \log\frac{1+x^2+2x}{1+x^2-2x} = \log\frac{(1+x)^2}{(1-x)^2}$$

$$= 2\log\frac{1+x}{1-x} = 2f(x)$$

Example 5.8

If $f(x) = x$ and $g(x) = |x|$, then find

(i) $(f+g)(x)$ (ii) $(f-g)(x)$ (iii) $(fg)(x)$

Solution

(i) $(f+g)(x) = f(x) + g(x) = x + |x|$

$$= \begin{cases} x+x & \text{if } x \geq 0 \\ x-x & \text{if } x < 0 \end{cases} = \begin{cases} 2x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

(ii) $(f-g)(x) = f(x) - g(x) = x - |x|$

$$= \begin{cases} x-x & \text{if } x \geq 0 \\ x-(-x) & \text{if } x < 0 \end{cases} = \begin{cases} 0 & \text{if } x \geq 0 \\ 2x & \text{if } x < 0 \end{cases}$$

(iii) $(fg)(x) = f(x)\,g(x) = x|x|$

$$= \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}$$

Example 5.9

A group of students wish to charter a bus for an educational tour which can accommodate atmost 50 students. The bus company requires at least 35 students for that trip. The bus company charges ₹ 200 per student up to the strength of 45. For more than 45 students, company charges each student ₹ 200 less $\frac{1}{5}$ times the number more than 45. Consider the number of students who participates the tour as a function, find the total cost and its domain.

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